Sanctions without Markups: Sourcing as the Margin of Bureaucratic Inefficiency*

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Abstract

Do sanctioned officials face higher prices because suppliers extract delivery-risk premia, or because the buyer's sourcing pattern shifts? The two predictions are observationally equivalent in cross-buyer data. Using bid-level data on 479,330 pharmaceutical purchases by the São Paulo State Department of Health (2009–2019), we exploit a parallel administrative-request channel that shares all planning constraints of court-mandated procurement but carries no penalty for officials. Manski-Lee bounds place the selection-corrected "under-the-gun" gap at $[15.9\%, 21.1\%]$. Within firm-buyer-item triples observed under both regimes, the Admin coefficient is statistically indistinguishable from zero ($\hat{\beta} = 0.035$, SE $= 0.041$): *there is no within-firm markup.* The price margin operates through demand fragmentation—admin orders are roughly 3.3× larger, mechanically delivering bulk discounts—and through equilib-

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rium changes in which firm wins. The bounded UTG implies a per-unit-price welfare cost of \$27.8 M per year on the \$300 M of annual litigated spending in São Paulo; the policy lever is demand aggregation, not contract design.
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1. Introduction

Do sanctioned officials face higher prices because suppliers extract delivery-risk premia, or because the buyer's sourcing pattern shifts? Standard models of bureaucratic accountability (Prendergast, 2007) predict the first; passive-waste accounts (Bandiera et al., 2009) predict the second. The two predictions are observationally equivalent in cross-buyer data, where buyer-supplier match heterogeneity confounds within-firm and between-firm margins. We provide a within firm-buyer-item test in a setting where accountability pressure should bite hardest: court-mandated pharmaceutical procurement in São Paulo under fines, personal liability, and fund seizure for delivery failure. The within-firm prediction fails; the sourcing prediction holds.

Setting and identification asset.. We use all pharmaceutical procurement by the São Paulo State Department of Health (SES/SP) from 2009 through 2019: 479,330 purchase-offer-item observations on the state's electronic procurement platform (BEC), which is the mandatory channel for common goods since 2007. The institutional asset that identifies our comparison is a parallel administrative-request channel, instituted by the SES/SP scientific committee in 2009: it shares all planning constraints of court-mandated procurement (timelines of 1–10 days, small quantities, diverted budget, identical auction procedures on BEC) but carries no penalty for officials when delivery fails. To our knowledge, no other Brazilian state operates a comparable dual channel for urgent pharmaceutical demand.

Findings.. First, there is no within-firm markup. Within firm-buyer-item triples observed under both regimes (1,206 triples, 4,573 observations), the Admin coefficient is $\hat{\beta} = 0.035$ (SE 0.041), statistically indistinguishable from zero. The same firm prices the same item to the same buyer at the same level under either regime; the delivery-risk-premium prediction is empirically rejected. *Second, the bounded under-the-gun gap is modest.* Under Manski-Lee monotone restrictions on committee selection, the gap lies in $[15.9\%, 21.1\%]$, at the lower end of within-buyer procurement-cost dispersion documented in comparable settings (e.g., Best et al., 2023; Bosio et al., 2022). *Third, the gap operates through demand fragmentation.* Administrative orders are roughly 3.3× larger than litigated orders; the bulk-discount channel mechanically delivers -32.8% in admin-minus-litigated log price. The within-firm null contributes zero, and the residual loads what the identity attributes neither to scale nor to pricing—an imprecisely identified sourcing-and-aggregation channel (Figure 1).

Identification discipline.. Selection into the administrative channel is non-random: the scientific committee admits on cost-effectiveness criteria. A pre-period probit confirms that admin over-represents items that would have been priced lower under any regime (coefficient on pre-period log reference price -0.051, $p < 0.001$; on SUS-formulary status -0.265, $p < 0.001$), motivating the Lee bound.[1] Wild cluster bootstrap on the preferred-FE Admin coefficient delivers $p_{\text{boot}} = 0.0080$ (999 Rademacher replicates, 95% CI in log-points $[-0.469, -0.039]$). A Borusyak-Jaravel-Spiess event-study with a Rambachan-Roth (Rambachan and Roth, 2023) honest sensitivity overlay is reported supportively in the Appendix; the cross-sectional FE design is primary. A placebo on items never subject to litigation delivers economically and statistically zero coefficients (-0.020, SE 0.032 for negotiated prices), confirming that the urgency premium does not exist outside the litigation pool.

Contributions, welfare, and literature.. The central contribution is conceptual: the sourcing-vs-pricing distinction. The cross-buyer literature cannot separate within-firm pricing from equilibrium supplier selection; the within firm-buyer-item triple does, and isolates sourcing under fragmentation as the empirical signature of accountability-induced inefficiency. Two supporting contributions follow. Whereas Coviello et al. (2018) document timeline *extension* as a procurement cost driver, we identify timeline *compression* under

[1] A parametric Heckman correction is non-identified in this design (Online Appendix A.19).

sanctions as an alternative channel, with the cost concentrated where demand aggregation matters most. The finding directly engages recent JPubE evidence on group purchasing in health (Lin and Wang, 2025), which identifies the same demand-aggregation lever we find compromised by court mandates. The paper engages procurement design and frictions (Bandiera et al., 2009; Best et al., 2023; Bosio et al., 2022; Decarolis et al., 2020; Carril et al., 2026), accountability-induced bureaucratic distortion (Prendergast, 2007; Bandiera et al., 2021), and the judicialization of health in Latin America (Wang, 2015; Ferraz, 2009; Biehl et al., 2009). Methodologically, the design honors post-treatment-bias (Acharya et al., 2016): quantity- and firm-fixed-effect controls are read as channel-isolating rather than direct-effect estimators. Applied to the \$300 M of annual litigated spending in São Paulo with an admissibility calibration of 50.0%, the bounded UTG implies a per-unit-price welfare cost of \$27.8 M per year (Lee range $[\$23.9\ M, \$31.7\ M]$). The policy lever is demand aggregation, not contract design.

2. Institutional Background

The setting has three layers: a constitutional right to health that generates mass litigation; a mandatory electronic procurement platform (BEC) that records every transaction at bid level; and a parallel administrative-request channel that since 2009 runs alongside the litigated channel under identical planning constraints but *without* personal-sanction exposure for the procurement officer. To our knowledge, no other Brazilian state and no Latin

American jurisdiction we have surveyed operates a comparable dual channel for urgent pharmaceutical demand.

2.1. Right to Health and Mass Litigation

Brazil's 1988 Constitution declares health "a right of all and a duty of the state" (Article 196). Brazilian courts interpret Article 196 strictly: any individual may obtain virtually any medication through litigation, from common analgesics to rare-disease treatments costing over US$400,000 per year.[2] Access is inexpensive—a prescription and a public defender suffice—and approximately 85% of first-instance claims are granted (CNJ/INSPER, 2019). Litigated health spending in São Paulo amounts to approximately US$300 million annually, about 5% of the state public health budget.

2.2. Sanctions and the Administrative-Request Channel

The São Paulo State Department of Health (SES/SP) delivers health services through 99 decentralized public buyer units (PBUs). All purchases pass through BEC, mandatory for common goods since 2007, with two competitive procedures: *convite* (sealed-bid, smaller lots) and *pregão eletrônico* (multi-round descending auction, dominant for urgent purchases).[3] Ordinary

[2]Key STF rulings: Recurso Extraordinário 855.178/SE (Tema 793, 2015), establishing joint-and-several liability of federal entities for health-service provision; and Recurso Extraordinário 566.471/RN (Tema 6, 2020) and Recurso Extraordinário 657.718/MG (Tema 500, 2019), defining conditions under which courts may order provision of high-cost and non-ANVISA medications respectively.

[3]Among the both-types cells that identify our tightest specifications, 95.5% of observations are processed through pregão eletrônico.

purchases use a planning phase of 30–180 days, allowing demand aggregation across PBUs. 99.94% of court decisions in our sample arrive as preliminary injunctions with delivery deadlines of 1–10 days, compressing the internal phase to roughly one-third of the ordinary timeline.

The penalties for non-compliance are concrete and personal. The binding sanction is a daily fine (*astreinte*) calibrated to the purchase value, frequently reaching the order value within weeks of accumulation; additional sanctions include administrative and criminal liability for the responsible official and sequestration of state funds. The court order is referenced in the tender notice, so suppliers see the sanction context as they bid.

The institutional asset that identifies our comparison is the *administrative-request channel*, instituted by the SES/SP in 2009. Patients (or social workers acting on their behalf) request a medication directly through SES/SP intake before any litigation; a scientific committee evaluates the request under evidence-based-medicine and cost-effectiveness criteria. Admitted requests proceed through urgent procurement on BEC under the same compressed-timeline regime as litigated purchases. The relevant asymmetry is in the penalty regime: **failure to complete an administrative purchase carries no court order, no daily fine, no fund seizure, and no personal liability.** The supplier pool is identical: 92% of urgent winners also serve ordinary procurement of the same item-class. Between 2009 and 2019, the administrative channel generated 9,700 purchases, about 6.5% of the litigated volume; the smaller volume reflects committee gating, not procedural difference. We bound the selection-into-admin wedge formally in Section 4.

Table 1 summarizes the three regimes.

Table 1: Purchase types: institutional characteristics.

	Ordinary	Administrative	Litigated
Trigger	Standard demand	Patient request	Court order
Gatekeeper	PBU plan	SES/SP committee	Lower-court judge
Source of funds	Dedicated budget	Diverted budget	Diverted budget
Quantity	Large	Small	Small
Delivery deadline	30–180 d	1–10 d	1–10 d
Penalty for failure	None	None	**Daily fine, liability, fund seizure**
Visible to bidders	—	—	Court order in tender notice
Active since	Pre-2007	2009	1990s

Notes: Administrative and litigated purchases share all execution constraints (compressed timeline, small quantity, diverted budget, BEC platform, identical auction procedures) and differ only in the penalty regime visible to the procurement officer and to bidders. The within-urgent contrast we exploit in Section 4 compares these two columns.

3. Data and Sample

3.1. Sources

We assemble three administrative datasets covering January 2009 through December 2019. The first is the universe of bid-level records from the BEC electronic procurement platform for SES/SP purchases of medical supplies (BEC Group 65); each observation is a purchase-offer-item (POI) recording the reference price set during the planning phase, the winning bid price, the number of bidding firms, the quantity ordered, the modality (*pregão eletrônico* or *convite*), and tender outcome. The second is a purchase-type classifier that assigns each POI to one of three regimes: ordinary, administrative-request, or litigated. By Brazilian law, the tender notice for any urgent purchase must reference the underlying court order or administrative-channel request; the classifier exploits this regulatory disclosure through a regular-expression algorithm applied to free-text tender notices. We validate the classifier through a stratified hand-labeling protocol over 500 notices ($\sim$167 per predicted class); detailed metrics are in the Online Appendix.[4] The third is the SES/SP S-CODES database of health-related litigation, from which we recover SUS-formulary status and injunction enforcement data for each litigated request.

The binary outcome *tender success* equals one if the POI closes with an accepted winning bid within its procurement order and zero otherwise; re-tendered items enter as new POIs and are not counted retroactively as successes. All continuous variables are winsorized at the 1st and 99th percentiles; robustness to alternative winsorization is in the Appendix.

3.2. Sample and descriptive patterns

The raw data comprise 479,330 POI observations across 33,144 distinct items, 51,059 purchase orders, and 100 PBUs (Online Appendix Table A.1). Our analysis sample restricts to items for which at least one ordinary and at least one litigated purchase is observed, ensuring within-item comparabil-

[4]The regex relies on literal court-order references whose presence or absence is independent of the price outcomes we estimate, so any residual misclassification attenuates rather than biases our coefficients.

ity; this yields 226,305 observations covering 3,856 items.[5] Price regressions further restrict to winning bids (196,995 observations).

Table 2 reports sample means by purchase type on winning bids. Three patterns anticipate the empirical results. *Higher litigated prices.* Log negotiated prices are higher in litigated than in ordinary purchases (1.050 vs. 1.596; item fixed effects absorb the level mix). *Larger administrative orders.* Administrative orders are markedly larger than litigated ones (794,534 vs. 61,903 units on average), the bulk-discount disadvantage that drives the mechanical channel in Section 5. *Thinner urgent competition with higher success.* The number of bidding firms is lower in urgent than in ordinary tenders (2.191 for litigated vs. 3.159 for ordinary), yet tender success rates are higher in urgent regimes (90.7% litigated vs. 84.1% ordinary)—officials accept worse terms to ensure compliance.

[5]Sample sizes vary slightly across tables due to missing values in specific dependent variables and the restriction to winners for price regressions.

Table 2: Sample means by purchase type.

	Ordinary	Administrative	Litigated
Observations (winners)	343,393	16,985	46,212
Log reference price	2.109	0.960	1.578
Log negotiated price	1.596	0.543	1.050
Quantity (units)	56,280	794,534	61,903
Number of bidding firms	3.159	2.394	2.191
Tender success rate (%)	84.1	85.7	90.7

Notes: Means computed on winners (POIs with an accepted winning bid). Reference and negotiated prices in logs; quantity in units; tender success rate is the share of POIs that close with an accepted winning bid. Sample period: 2009–2019. Source: BEC-SP and SES/SP S-CODES.

3.3. The under-the-gun subsample

For the under-the-gun analysis, we construct a subsample of urgent purchases (administrative and litigated) for items with both regimes present, yielding 56,803 observations (Online Appendix Table A.2). The tightest specifications identify off item×year-month cells in which both regimes coexist: of 40,966 such cells, 4,402 (11%) carry within-cell variation. The remaining cells are absorbed as singletons under the fixed-effects design. Both-types cells are not a representative subsample—they tilt toward higher-volume, more-reordered medications (Online Appendix Table A.20); the bounded estimate therefore speaks to the more liquid items in the urgent panel, a restriction we revisit in the welfare bound. Online Appendix complements the balance with geographic evidence on the spatial distribution of administrative demand, which mirrors the litigation patterns of Section 2.

4. Empirical Strategy

We exploit two within-item contrasts. The urgent-vs-ordinary contrast identifies the cost margin associated with the urgent regime, conditional on item, time, and PBU fixed effects. The under-the-gun (UTG) contrast, restricted to the urgent panel, identifies the cost margin associated with the sanction regime, conditional on selection into the administrative channel that we bound formally.

4.1. Specifications

The urgent-vs-ordinary premium is

$$y_{i,g,t} = \beta \cdot \mathrm{Urgent}_{i,g,t} + \gamma_g + \lambda_t + \delta_b + \varepsilon_{i,g,t}, \tag{1}$$

where $y_{i,g,t}$ is log negotiated price, log reference price, log number of firms, or the success indicator for purchase order i of item g at time t; γ_g, λ_t, δ_b are item, time, and PBU fixed effects. We report four specifications with progressively richer fixed effects. The under-the-gun coefficient restricts to the urgent panel (admin or litigated):

$$\ln(\mathrm{Price}_{i,g,t}) = \beta \cdot \mathrm{Admin}_{i,g,t} + \gamma_g + \lambda_t + \delta_b + \varepsilon_{i,g,t}, \tag{2}$$

estimated as (i) a naive coefficient, (ii) a Manski-Lee bounded coefficient, and (iii) a within firm-buyer-item triple coefficient that isolates same-firm pricing from supplier composition. Standard errors cluster at the PBU level

(99 clusters), with wild cluster bootstrap as a robustness check.

4.2. Identification: urgent vs. ordinary

Three institutional features support strict exogeneity of urgency conditional on item-time-PBU fixed effects. First, court orders and administrative requests originate from patient health needs and external (legal or scientific-committee) assessments—not from the procurement process. Second, urgent demand is poorly correlated with PBU mismanagement: only 4% of claims arise from stock shortages or service failures. Third, the urgency regime is determined externally, not by the purchasing unit. We frame the estimate as a within-item cost margin associated with the urgent regime, conditional on the institutional features above.

A Borusyak-Jaravel-Spiess (2024) imputation event-study around each item's first court order provides a dynamic check (Online Appendix). Pre-period coefficients are economically small (max absolute pre-period coefficient $= 0.041$); post-period coefficients rise from 0.052 at $t = 0$ to 0.147 at $t = +5$. A Rambachan-Roth (2023) honest sensitivity test gives a breakdown linear-extrapolation parameter at which the $t = 0$ confidence interval just contains zero of $M = 0.018$. The cross-sectional FE estimate is the primary object; the BJS event-study is supportive dynamic evidence rather than a stand-alone identification claim.

4.3. Identification: under-the-gun

Litigated and administrative urgent purchases face identical planning constraints (Section 2) and differ only in penalty exposure. Selection into the administrative channel is non-random: a scientific committee admits cases on cost-effectiveness criteria. The committee's first-stage probit on pre-period item characteristics rejects higher pre-period log reference price (-0.051, $p < 0.001$) and SUS-formulary status (-0.265, $p < 0.001$): admin systematically over-represents items that would have been priced lower under any regime, so the naive UTG comparison inflates the sanction effect.

We bound this wedge under monotone selection (Lee 2009) by trimming admin observations within item×year×PBU strata where the admin sample exceeds the litigated count. The trimming proportion $p_{\text{trim},s} = \max\{0, (n_{\text{adm},s} - n_{\text{lit},s})/n_{\text{adm},s}\}$ is the share of admin observations within stratum s that, under the Lee monotone-acceptance assumption, would not have appeared in the litigated counterfactual. Trimming admin from the upper tail of the within-stratum price distribution yields the lower bound on the admin-minus-lit log-price gap; trimming from the lower tail yields the upper bound. Average trimming intensity is 26.9% of admin observations where applicable; 12,038 of 44,654 strata trigger trimming. A parametric Heckman correction is non-identified in this design (Online Appendix A.19); the Lee bound is the primary selection-corrected object.

We do not estimate a structural counterfactual price under hypothetical sanction removal. That object is unidentified in our setting because the administrative subsample is selected on item economics. What we provide is a bounded decomposition of where the between-regime cost margin originates.

5. Results

5.1. Three motivating patterns

Within the same item, urgent purchases carry negotiated prices 5.4% higher than ordinary purchases and attract -5.4% fewer bidding firms; reference prices set during the planning phase rise by 2.7% ($p < 0.10$), attenuating sharply once buyer fixed effects absorb buyer-level heterogeneity. Despite worse conditions, urgent tenders *succeed* more often: 2.1 pp ($p < 0.01$). Officials accept worse terms to ensure compliance rather than cancel; the urgency premium does not arise because urgency selects easier purchases.[6]

[6] The 5.4% within-item premium sits at the lower end of within-buyer procurement-cost dispersion documented in comparable settings: Bandiera-Prat-Valletti 2009 10–20%; Best, Hjort & Szakonyi 2023 20–40%; Bosio et al. 2022 15–25%; Coviello-Guglielmo-Spagnolo 2018 5–12%. Tighter within-buyer within-item identification is consistent with a smaller headline number.

Table 3: Negotiated Prices

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Urgent Purchase	0.159***	0.151***	0.053***	0.056***
	(0.042)	(0.038)	(0.016)	(0.016)
Panel B: Direct Effect				
Urgent Purchase	0.075	0.060	0.030*	0.030*
	(0.045)	(0.042)	(0.017)	(0.017)
Log Quantity	-0.321***	-0.325***	-0.392***	-0.393***
	(0.026)	(0.026)	(0.029)	(0.029)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,886	196,886	196,883	196,883
Within R^2 (A)	0.003	0.003	0.000	0.000
Within R^2 (B)	0.315	0.327	0.358	0.359

Notes: Dependent variable: log negotiated price. Sample: winners only. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

5.2. Same firm, same buyer, same item: no within-firm markup

Restricting to firm-buyer-item triples observed under both the litigated and the administrative regime (1,206 triples covering 4,573 observations), the Admin coefficient is 0.035 (SE 0.041): statistically indistinguishable from zero (95% bootstrap CI [-6.6%, 11.0%]). The same firm prices the same item to the same buyer at the same level under either regime; suppliers who already serve the buyer's urgent procurement do not extract a markup when the official is sanction-exposed.

This is the empirical pivot of the paper. The standard accountability-distortion intuition (Prendergast, 2007) predicts that sanction-exposed officials face higher per-unit prices because suppliers extract a delivery-risk premium; in our data, that mechanism is absent. Whatever distinguishes litigated from administrative urgent procurement, it is not opportunistic pricing by incumbent suppliers.

5.3. The gap is sourcing, not pricing: decomposition and heterogeneity

Decompose the observed log-price gap (admin minus litigated, -0.259) into an arithmetic identity:

$$\underbrace{\Delta y_{\text{obs}}}_{-0.259} = \underbrace{\beta_{\text{bulk}} \cdot \Delta \ln Q_{\text{adm-lit}}}_{\text{mechanical: } -0.397} + \underbrace{\beta_{\text{within-firm}}}_{\text{within-firm: } +0.035} + \underbrace{\beta_{\text{composition}}}_{\text{residual: } +0.103} .$$

Three readings follow, with 95% cluster-bootstrap CIs (PBU, 499 replicates). *Bulk-discount channel identified and large*: -32.8% ([-51.2%, -5.2%]). Order-size differences alone would make admin cheaper by 32.8%, more than the 23% actually observed. *Within-firm channel identified and zero*: 3.5% ([-6.6%, 11.0%]). *Residual channel imprecisely identified*: 10.9% on the point estimate, ([-16.0%, 39.6%]) on the bootstrap, and inherits variance from the other three bars by construction. The residual loads what the identity attributes neither to scale nor to within-firm pricing: equilibrium changes in *which* firm wins, match-quality shifts, and any other margin not separately identified in this design.

Table 4: UTG reconciliation: observed log-price gap decomposed into mechanical bulk-discount prediction, within firm-buyer-item offset, and supplier composition residual. 1,206 triples observed in both regimes.

Component	Log-points (admin minus lit)
Observed UTG (admin minus lit, log price)	-0.259
Mechanical C1: bulk-discount x qty gap	-0.397
Within firm-buyer-item offset	0.035
Supplier composition residual	0.104

Identity: observed UTG = mechanical C1 + within-firm offset + composition residual. Within-fi

Where the cost bites confirms the sourcing reading.. Splitting items by mean number of participating firms (median 2.071), competitive items show a 14.4% urgency premium with 8.5% surviving firm fixed effects (39% attenuation). Concentrated items show no detectable premium (-1.0%, not significant). Urgency erodes competitive pressure where there is competitive pressure to erode—the heterogeneity matches the sourcing channel, not a pricing channel.

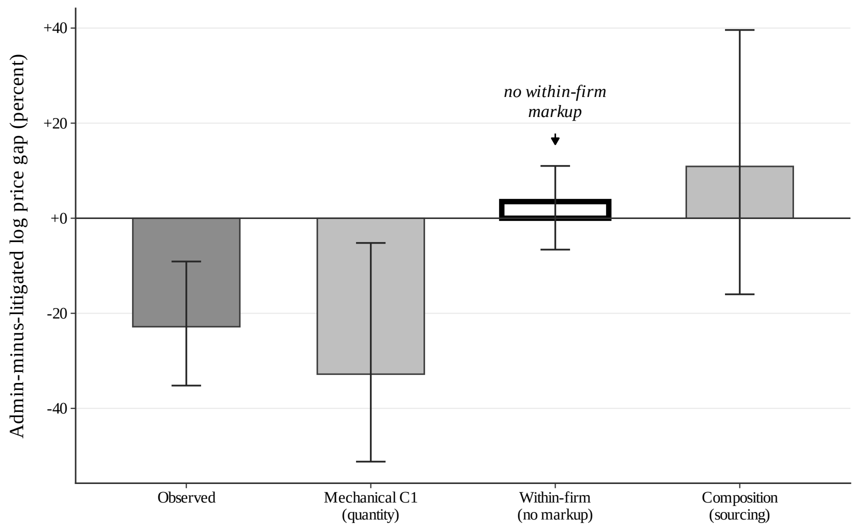

Figure 1: Sourcing vs. pricing: the within-firm null is the empirical pivot.

Notes: Bars give the four components of the admin-minus-lit log-price identity in Table A.22; whiskers are 95% cluster bootstrap on PBU with 499 replicates. The within firm-buyer-item bar (white, heavy outline) brackets zero: same firm, same buyer, same item, same price across regimes. The mechanical bulk-discount channel is identified and over-explains the observed gap; the residual composition channel loads the remainder but is imprecisely identified.

5.4. Selection bounds on the size of the gap

The naive Admin coefficient on the urgent-only subsample is -0.262 ($p < 0.05$) under the preferred FE, implying litigated purchases are 29.5% more expensive than administrative ones; wild cluster bootstrap (Rademacher, $B = 999$) confirms with $p = 0.0080$ and 95% CI in log-points $[-0.469, -0.039]$. This comparison is contaminated by selection into the administrative channel: the SES/SP scientific committee admits cases on cost-effectiveness criteria. The committee's selection rule, recovered from a probit on pre-period item characteristics (N = 5,351), rejects higher pre-period log reference prices (-0.051, $p < 0.001$) and SUS-formulary status (-0.265, $p < 0.001$);

Table 5: Manski-Lee bounds on the under-the-gun coefficient.

Specification	log negotiated price Coef. on Admin	SE	implied Lit-vs-Admin (%)
Naive UTG (item + year + PBU FE)	-0.259	0.092	29.530
Lee lower bound (admin top-trim)	-0.192	0.095	15.924
Lee upper bound (admin bottom-trim)	-0.148	0.097	21.123

Trimming applied within item*times*year*times*PBU strata where the admin sample exceeds the li

admin over-represents items that would have been priced lower under any regime.

We bound the selection wedge using Lee (2009) trimming within item×year×PBU strata where the admin sample exceeds the litigated count. Trimming admin from the upper tail of the within-stratum price distribution gives the lower bound; trimming from the lower tail gives the upper bound.

The bounded gap falls in $[15.9\%, 21.1\%]$, well below the naive 29.5%: selection on observables explains approximately one-third of the headline gap. The within-cell variation that identifies the tightest specification comes from a non-representative subsample (Online Appendix Table A.20): of 40,966 item×month cells, only 5,581 (about 11%) carry within-cell variation and they tilt toward higher-volume, more-frequent SUS-formulary medications. The bounded estimate therefore speaks to the more liquid items in the urgent panel, not to the universe of court-mandated purchases.

5.5. Falsification: items never litigated

Running the preferred specification on items with zero litigated purchases across 2009–2019 (restricted to those with both ordinary and administrative purchases) yields placebo coefficients of -0.020 (SE 0.032) for negotiated prices and -0.031 (SE 0.045) for reference prices. Both are economically small and statistically insignificant (Online Appendix Table A.8). The urgency premium exists only for items that are targets of litigation.

6. Conclusion

Courts that compel governments to buy medications save individual lives. They also raise public procurement costs by 5.4% per purchase across the full sample, attract -5.4% fewer bidders, and—once selection into the SES/SP administrative channel is bounded—generate an under-the-gun gap of $[15.9\%, 21.1\%]$. The empirical signature of this inefficiency in our data is the *absence* of a within-firm markup, not its presence: within firm-buyer-item triples observed under both the litigated and the administrative regime, the same firm prices the same item to the same buyer at a statistically indistinguishable level (3.5%, bootstrap CI [-6.6%, 11.0%]). Suppliers do not extract a markup when the official is sanction-exposed. The gap is sourcing under fragmentation, not pricing under sanctions.

Welfare bound.. Applied to \$300 M of annual litigated spending in São Paulo and an admissibility calibration of 50.0%, the bounded UTG implies a per-unit-price welfare cost of \$27.8 M per year (Lee range $[\$23.9\ M, \$31.7\ M]$).

This is a per-unit-price margin and excludes welfare losses from constrained order sizes (the bulk-discount channel itself), reduced bidder participation, and officials' search time diverted from ordinary procurement.

Policy implications.. Two reforms address the two channels we identify. *Demand side*: expanding the administrative-request mechanism eliminates within-urgent sanction exposure for the admissible share. Because the within-firm pricing component is zero, the gain is realized through demand aggregation; the binding parameter is committee capacity, not legal reform. *Supply side*: framework agreements for repeat-purchase commodity-like medications address fragmentation directly by allowing aggregation across urgent and ordinary requests, the instrument introduced under Lei 14.133/2021. We do not attach a recovery point estimate to the supply-side channel: any number requires a calibration anchor on item eligibility we are not positioned to provide.

Boundary and outlook.. Results are estimated within São Paulo's BEC platform, on pharmaceutical procurement in a single state, in 2009–2019. We do not extrapolate to other states, to non-pharmaceutical procurement, or to post-reform institutional arrangements. The bounded estimate applies within the urgent panel and tilts toward higher-volume formulary medications (Online Appendix Table A.20); the welfare figure assumes that admin admissibility can be expanded without affecting the per-unit-price gap, an assumption the manuscript does not test. The broader implication is methodological: standard accountability designs are rationalized on the prediction that sanctions discipline prices through suppliers' delivery-risk premia. The within firm-buyer-item null we recover here suggests that, where the binding margin is sourcing rather than pricing, the policy lever shifts from contract design to demand aggregation.

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Online Appendix

This appendix collects tables and figures from the main text and robustness checks. Section A.0 reports baseline descriptives and balance. Section A.1 examines winsorization sensitivity. Section A.2 traces the under-the-gun coefficient under progressive controls. Section A.3 plots outcome distributions by purchase type. Section A.4 maps administrative-channel demand. Section A.5 reports the BJS event study with Rambachan-Roth honest sensitivity. Section A.6 examines heterogeneity. Section A.7 documents the regex classifier validation. Sections A.8–A.9 collect the Heckman, both-types-cell, wild bootstrap, and reconciliation tables referenced from the main text.

A.0 Supporting Descriptives, Balance, and Secondary Outcome Tables

Table A.1: Descriptive Statistics by Purchase Type

	Ordinary		Administrative		Litigated		Diff	Diff
	Mean	(SD)	Mean	(SD)	Mean	(SD)	(O–L)	(A–L)
Panel A: Levels								
Reference Price	909.17	(5,959.41)	226.38	(2,257.35)	370.39	(2,540.20)	538.78***	-144.01***
							[0.000]	[0.000]
Negotiated Price	477.51	(2,915.88)	148.03	(1,177.52)	260.25	(1,561.81)	217.26***	-112.22***
							[0.000]	[0.000]
Quantity	31,487.15	(163,507.29)	117,146.00	(356,962.37)	9,646.19	(82,683.10)	21,840.96***	107,499.81***
							[0.000]	[0.000]
N. Bidding Firms	3.17	(1.93)	2.43	(1.35)	2.20	(1.19)	0.97***	0.23***
							[0.000]	[0.000]
Panel B: Log Transformations								
Log Reference Price	0.926	(2.723)	0.909	(2.541)	1.557	(2.446)	-0.632***	-0.648***
							[0.000]	[0.000]
Log Negotiated Price	0.319	(2.862)	0.480	(2.604)	1.024	(2.568)	-0.705***	-0.544***
							[0.000]	[0.000]
Log Quantity	6.947	(2.678)	7.472	(3.160)	6.087	(2.083)	0.860***	1.385***
							[0.000]	[0.000]
Log N. Firms	0.984	(0.590)	0.747	(0.529)	0.665	(0.493)	0.319***	0.081***
							[0.000]	[0.000]
Panel C: Tender Characteristics								
Successful Tender (%)	0.859	(0.348)	0.869	(0.338)	0.908	(0.288)	-0.050***	-0.040***
							[0.000]	[0.000]
Observations	136,250		15,371		45,351			

Notes: Sample restricted to items with at least one ordinary and one litigated purchase. Winners only for price and quantity variables. Variables winsorized at 1%/99%. Stars on differences from Welch t-tests; p-values in brackets. *** p<0.01, ** p<0.05, * p<0.1.

Table A.2: Balance Table: Administrative vs Litigated (Urgent Purchases)

	Administrative		Litigated		Diff	p-value
	Mean	(SD)	Mean	(SD)	(A–L)	
Panel A: Procurement Outcomes						
Reference Price	226.38	(2,257.35)	232.56	(1,432.18)	-6.18	[0.752]
Negotiated Price	148.03	(1,177.52)	174.79	(1,052.56)	-26.75**	[0.013]
Quantity	117,146.00	(356,962.37)	9,306.87	(80,252.25)	107,839.13***	[0.000]
Log Reference Price	0.91	(2.54)	1.43	(2.32)	-0.52***	[0.000]
Log Negotiated Price	0.48	(2.60)	0.89	(2.44)	-0.41***	[0.000]
Log Quantity	7.47	(3.16)	6.18	(1.99)	1.29***	[0.000]
Panel B: Market Structure						
N. Bidding Firms	2.427	(1.351)	2.155	(1.120)	0.272***	[0.000]
Log N. Firms	0.747	(0.529)	0.650	(0.483)	0.097***	[0.000]
Successful Tender (%)	0.869	(0.338)	0.914	(0.280)	-0.045***	[0.000]
Panel C: Purchase Characteristics						
Electronic Auction	0.985	(0.123)	0.941	(0.236)	0.044***	[0.000]
Observations	15,371		41,491			

Notes: Sample restricted to urgent purchases (administrative and litigated) for items with both types present. Winners only for price/quantity. Variables winsorized at 1%/99%. p-values from Welch t-tests in brackets. *** p<0.01, ** p<0.05, * p<0.1.

Table A.3: Within-Item Balance: Administrative vs. Litigated

Covariate	Mean (Admin)	Mean (Lit)	Raw diff. (A–L)	Within-item diff. Coef	Within-item diff. (SE)
SUS basic (MEDICAMENTO flag)	0.718	0.892	-0.174	0.000^{NA}	(NaN)
Electronic auction (pregão)	0.984	0.941	0.043	0.041^{**}	(0.017)
Log quantity (bulk size signal)	7.642	6.180	1.462	0.866^{*}	(0.502)
Log reference price	0.835	1.401	-0.566	-0.339^{***}	(0.093)
Successful tender	0.869	0.914	-0.045	-0.024	(0.022)
Late period (year $\geq$ 2014)	0.565	0.623	-0.058	-0.045	(0.041)
Large PBU (above-median)	0.895	0.810	0.085	0.083	(0.073)
Observations			63,426		
Items			2,370		

Notes: Within-item balance between administrative ($Admin = 1$) and litigated ($Admin = 0$) urgent purchases, restricted to items with both types present. Raw difference is the unconditional mean gap. Within-item coefficient is β from $x_{i,g,t} = \beta\,\text{Admin}_{i,g,t} + \gamma_g + \varepsilon_{i,g,t}$, where γ_g is an item fixed effect and standard errors are clustered at the PBU level. A coefficient close to zero means that, within a given item, the administrative and litigated draws are balanced on the covariate. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A.4: Reference Prices

	(1)	(2)	(3)	(4)
Urgent Purchase	0.164***	0.156***	0.027*	0.027*
	(0.056)	(0.050)	(0.014)	(0.014)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896
Within R^2	0.003	0.003	0.000	0.000

Notes: Dependent variable: log reference price. Sample: winners only, items with both ordinary and litigated purchases. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.5: Quantities

	(1)	(2)	(3)	(4)
Urgent Purchase	-0.264	-0.279	-0.058	-0.065
	(0.183)	(0.181)	(0.056)	(0.058)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896
Within R^2	0.003	0.003	0.000	0.000

Notes: Dependent variable: log quantity. Sample: winners only. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

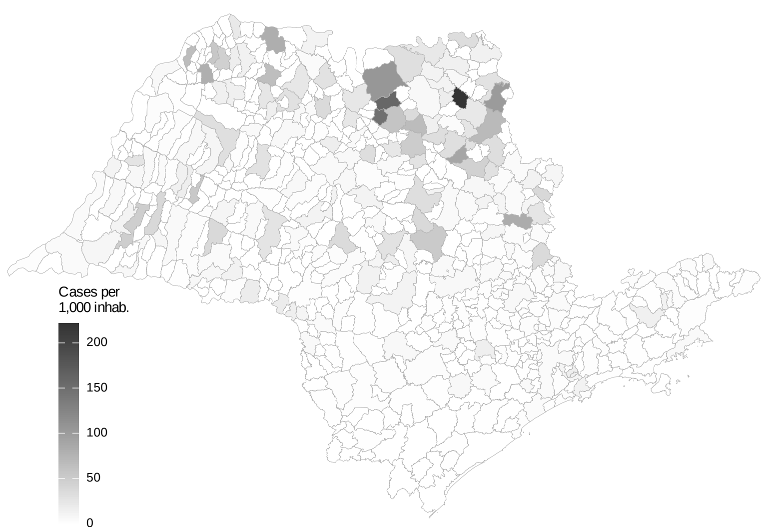


Figure A.1: Health Litigation Cases per 1,000 Inhabitants Across Municipalities in São Paulo (2009–2019)

Source: Authors' elaboration based on CNJ/INSPER (2019) litigation data and IBGE population estimates.

Table A.6: Participant Firms

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Urgent Purchase	-0.101***	-0.113***	-0.056***	-0.054***
	(0.023)	(0.022)	(0.014)	(0.015)
Panel B: Direct Effect				
Urgent Purchase	-0.091***	-0.102***	-0.042***	-0.040***
	(0.014)	(0.011)	(0.012)	(0.013)
Log Quantity	0.063***	0.064***	0.065***	0.065***
	(0.004)	(0.004)	(0.003)	(0.003)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	81,805	81,805	81,801	81,801
Obs. (Panel B)	81,793	81,793	81,789	81,789
Within R^2 (A)	0.004	0.006	0.001	0.001
Within R^2 (B)	0.063	0.069	0.043	0.044

Notes: Dependent variable: log number of bidding firms. Sample: winners only. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A.7: Tender Success (LPM)

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Urgent Purchase	0.023**	0.023**	0.021***	0.021***
	(0.009)	(0.009)	(0.006)	(0.006)
Panel B: Direct Effect				
Urgent Purchase	0.023**	0.023**	0.021***	0.022***
	(0.009)	(0.010)	(0.006)	(0.006)
Log Quantity	0.003	0.003	0.012***	0.012***
	(0.004)	(0.004)	(0.002)	(0.002)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	226,305	226,305	226,301	226,301
Obs. (Panel B)	226,282	226,282	226,278	226,278
Within R^2 (A)	0.001	0.001	0.000	0.000
Within R^2 (B)	0.001	0.001	0.004	0.004

Notes: Dependent variable: successful tender (LPM). Sample: all observations. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.8: Placebo Test: Items Never Subject to Litigation

	bid_price_log		bid_price_ref_log	
	(1)	(2)	(3)	(4)
Urgent Purchase	-0.0204	0.0513***	-0.0306	0.0305**
	(0.0316)	(0.0153)	(0.0447)	(0.0141)
Observations	39,283	196,883	46,874	226,278
R^2	0.83364	0.86796	0.81121	0.85183
Within R^2	7.03×10^{-6}	0.00022	1.73×10^{-5}	7.43×10^{-5}
item_id fixed effects	✓	✓	✓	✓
year_n fixed effects	✓	✓	✓	✓
pbu_id fixed effects	✓	✓	✓	✓

Placebo sample: items with zero litigated purchases across 2009-2019. Main sample: items with at least one litigated and one ordinary purchase. DV: log price. Item + Year + PBU FE. SE clustered at PBU level.

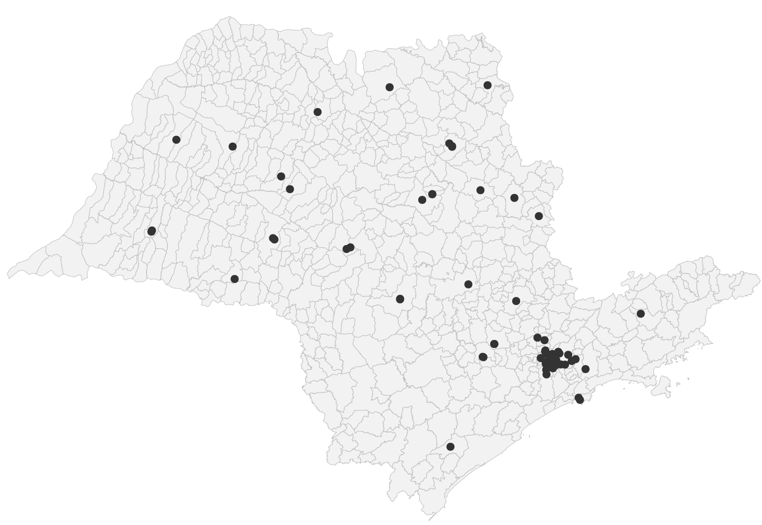


Figure A.2: Public Buyer Units (SES/SP)

Source: Authors' elaboration based on SES/SP administrative records.

A.1 Winsorization Sensitivity

Our baseline estimates use 1%/99% winsorization to limit the influence of extreme outliers while preserving meaningful variation in the data. To verify that our results are not driven by this specific choice, Tables A.9–A.12 replicate the main regression tables under three alternative winsorization treatments: no winsorization (Panel A), the 1%/99% baseline (Panel B), and a more aggressive 5%/95% winsorization (Panel C).

Table A.9 reports the sensitivity of the reference price estimates. The positive coefficient on urgent purchases is stable across all three winsorization levels: the preferred specification (column 3) yields estimates of 0.026 (no winsorization), 0.027 (baseline), and 0.027 with aggressive winsorization. The qualitative pattern—a large raw premium that attenuates substantially with PBU fixed effects—is preserved throughout, indicating that the reference price result is not driven by outliers.

Table A.10 presents the corresponding sensitivity analysis for negotiated prices. The total effect of urgency on negotiated prices is positive and statistically significant across all winsorization levels. The coefficient magnitudes are slightly larger without winsorization (reflecting the influence of extreme values) and slightly smaller with aggressive winsorization, but the economic message remains unchanged: urgent purchases are associated with higher negotiated prices, with the preferred estimate ranging from approximately 5% to 6%.

Table A.11 examines the robustness of the bidder participation results.

Table A.9: Robustness: Reference Prices

	(1)	(2)	(3)	(4)
Panel A: No Winsorization				
Urgent Purchase	0.167***	0.159***	0.026*	0.026*
	(0.056)	(0.051)	(0.014)	(0.014)
Panel B: Winsorized 1%/99%				
Urgent Purchase	0.164***	0.156***	0.027*	0.027*
	(0.056)	(0.050)	(0.014)	(0.014)
Panel C: Winsorized 5%/95%				
Urgent Purchase	0.156***	0.150***	0.027*	0.028**
	(0.053)	(0.048)	(0.014)	(0.014)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896

Notes: Dependent variable: log reference price. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

The negative effect of urgency on the number of bidding firms is remarkably stable across winsorization levels, with preferred estimates in the range of -0.06 to -0.05 log points. This stability is expected, since the number of firms is a count variable with limited extreme values, making it less sensitive to winsorization.

Table A.12 replicates the tender success analysis. The positive coefficient on urgency—indicating that urgent tenders are more likely to succeed—is robust across all winsorization levels. The magnitude is consistent at approximately 2 pp in the preferred specification, reinforcing the interpretation

Table A.10: Robustness: Negotiated Prices

	(1)	(2)	(3)	(4)
Panel A: No Winsorization				
Urgent Purchase	0.160***	0.152***	0.051***	0.054***
	(0.043)	(0.038)	(0.015)	(0.016)
Panel B: Winsorized 1%/99%				
Urgent Purchase	0.159***	0.151***	0.053***	0.056***
	(0.042)	(0.038)	(0.016)	(0.016)
Panel C: Winsorized 5%/95%				
Urgent Purchase	0.162***	0.153***	0.058***	0.061***
	(0.041)	(0.037)	(0.017)	(0.017)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,886	196,886	196,883	196,883

Notes: Dependent variable: log negotiated price. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

that procurement officials under pressure accept less favorable terms to ensure compliance.

Table A.11: Robustness: Participant Firms

	(1)	(2)	(3)	(4)
Panel A: No Winsorization				
Urgent Purchase	-0.101***	-0.113***	-0.056***	-0.054***
	(0.023)	(0.022)	(0.015)	(0.015)
Panel B: Winsorized 1%/99%				
Urgent Purchase	-0.101***	-0.113***	-0.056***	-0.054***
	(0.023)	(0.022)	(0.014)	(0.015)
Panel C: Winsorized 5%/95%				
Urgent Purchase	-0.100***	-0.111***	-0.055***	-0.053***
	(0.022)	(0.021)	(0.014)	(0.015)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (A)	81,377	81,377	81,373	81,373
Obs. (B)	81,805	81,805	81,801	81,801
Obs. (C)	81,805	81,805	81,801	81,801

Notes: Dependent variable: log number of bidding firms. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.12: Robustness: Tender Success (LPM)

	(1)	(2)	(3)	(4)
Panel A: No Winsorization				
Urgent Purchase	0.023**	0.023**	0.021***	0.021***
	(0.009)	(0.009)	(0.006)	(0.006)
Panel B: Winsorized 1%/99%				
Urgent Purchase	0.023**	0.023**	0.021***	0.021***
	(0.009)	(0.009)	(0.006)	(0.006)
Panel C: Winsorized 5%/95%				
Urgent Purchase	0.023**	0.023**	0.021***	0.021***
	(0.009)	(0.009)	(0.006)	(0.006)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	226,305	226,305	226,301	226,301

Notes: Dependent variable: successful tender (LPM). Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

A.2 Under the Gun: Progressive Controls

Table A.13 examines the sensitivity of the "under the gun" estimate to the progressive inclusion of control variables and to the choice of winsorization level. Starting from the baseline specification with item + year + PBU fixed effects (column 1), we sequentially add log quantity (column 2), log reference price (column 3), log number of firms (column 4), and finally replace year FE with year-month FE (column 5). Each row within a panel shows how the coefficient on the administrative indicator responds to additional controls.

The key finding is that the administrative indicator remains negative across all specifications and winsorization levels, indicating that litigated purchases are consistently more expensive than administrative ones. The coefficient is largest and most precisely estimated in the baseline specification without additional controls (column 1), where it captures the total effect including indirect channels through quantity and competition. As controls are added, the coefficient attenuates—consistent with some of the price premium operating through the quantity and competition channels—but the sign is preserved throughout. This pattern confirms that the sanction channel has an economically meaningful direct effect on prices beyond its indirect effects through procurement conditions.

Table A.13: Robustness: Under the Gun with Progressive Controls

	(1)	(2)	(3)	(4)	(5)
Panel A: No Winsorization					
Administrative (vs Litigated)	-0.259**	0.138	0.115**	0.201**	0.195**
	(0.095)	(0.130)	(0.054)	(0.090)	(0.087)
Panel B: Winsorized 1%/99%					
Administrative (vs Litigated)	-0.262**	0.117	0.110*	0.199**	0.194**
	(0.096)	(0.122)	(0.055)	(0.093)	(0.090)
Panel C: Winsorized 5%/95%					
Administrative (vs Litigated)	-0.261**	0.060	0.089*	0.169*	0.165*
	(0.096)	(0.093)	(0.049)	(0.084)	(0.080)
Log Quantity	No	Yes	Yes	Yes	Yes
Log Ref. Price	No	No	Yes	Yes	Yes
Log N. Firms	No	No	No	Yes	Yes
Item FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	No
Year-Month FE	No	No	No	No	Yes
PBU FE	Yes	Yes	Yes	Yes	Yes
Obs. (A)	56,803	56,803	56,803	14,913	14,913
Obs. (B)	56,803	56,803	56,803	15,052	15,052
Obs. (C)	56,803	56,803	56,803	15,052	15,052

Notes: Dependent variable: log negotiated price. UTG sample (urgent, winners, items with both administrative and litigated). Progressive controls added left to right. Column (5) replaces Year with Year-Month FE. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

A.3 Distributional Evidence

The figures below show kernel density estimates by purchase type for the four main outcomes; the distributional shifts are consistent with the average effects estimated in the main text.

Figure A.3 (log reference prices): the litigated distribution is shifted right relative to ordinary; administrative lies in between.

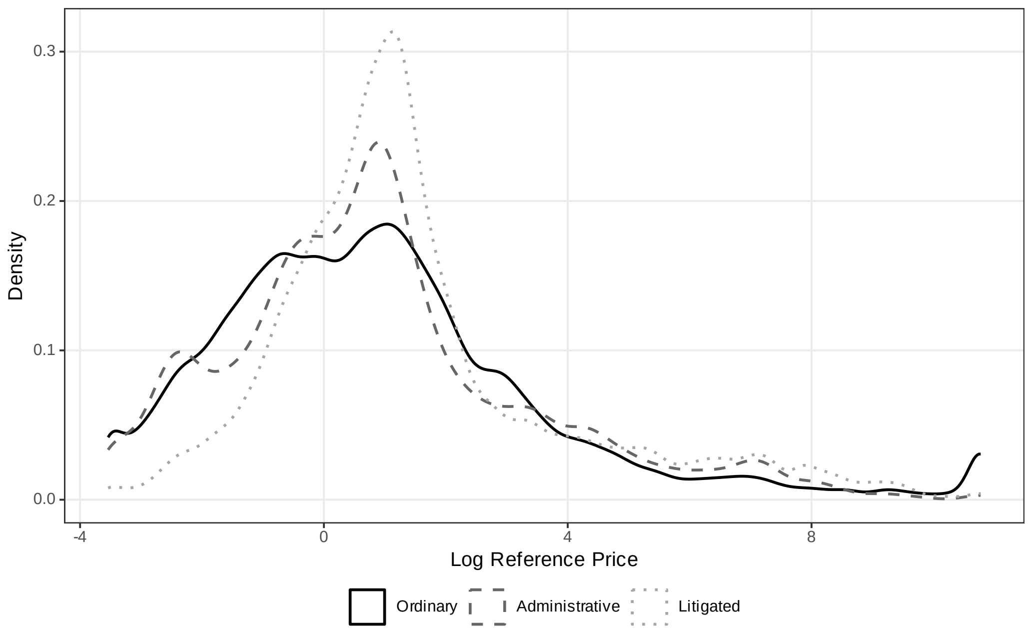


Figure A.3: Distribution of Log Reference Price by Purchase Type

Figure A.4 (log negotiated prices): same direction of shift; litigated has a thicker right tail.

Figure A.5 (log quantities): litigated shifts *left*; ordinary has a heavier right tail (demand aggregation under standard timelines); administrative is bimodal, spanning individual-prescription and stock-replenishment scales.

Figure A.6 (log number of bidding firms): litigated shifts left, with mass at low participation levels.

Figure A.7 focuses on the "under the gun" subsample, comparing the distributions of log negotiated prices for administrative and litigated urgent purchases only. The rightward shift of the litigated distribution relative to administrative purchases provides direct visual evidence of the sanction

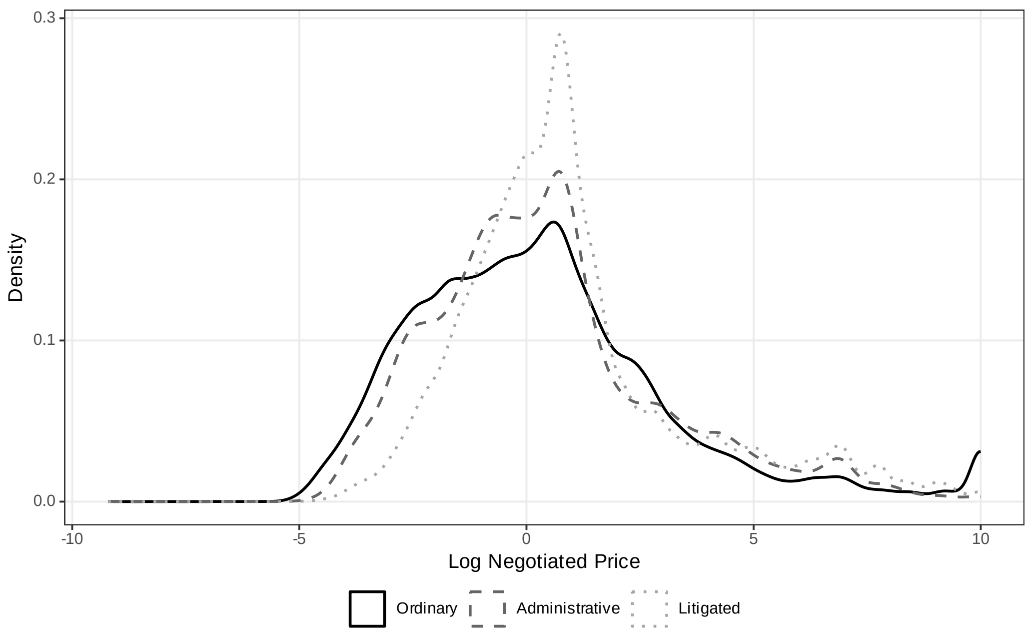


Figure A.4: Distribution of Log Negotiated Price by Purchase Type

channel: holding constant the urgency of the purchase, judicial pressure shifts the entire price distribution to the right.

Figure A.8 compares tender success rates across purchase types. Consistent with the LPM estimates, litigated purchases have the highest success rate, followed by administrative and ordinary purchases. This ordering is consistent with the "under the gun" mechanism: the greater the penalty for procurement failure, the higher the observed success rate, as officials accept less favorable terms to avoid sanctions.

Figure A.9 plots mean log negotiated prices over time for ordinary and urgent purchases separately. The figure reveals two patterns. First, the price gap between urgent and ordinary purchases is persistent throughout

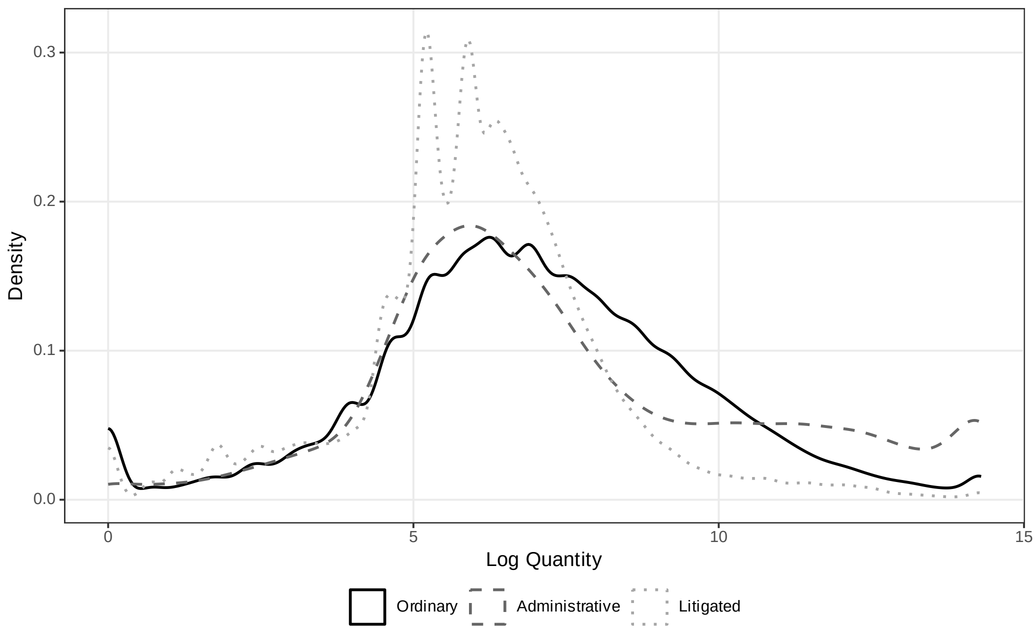


Figure A.5: Distribution of Log Quantity by Purchase Type

the sample period, ruling out the possibility that the regression estimates are driven by a specific subperiod. Second, both series exhibit common trends, supporting the identification assumption that time-varying confounders do not differentially affect the two purchase types.

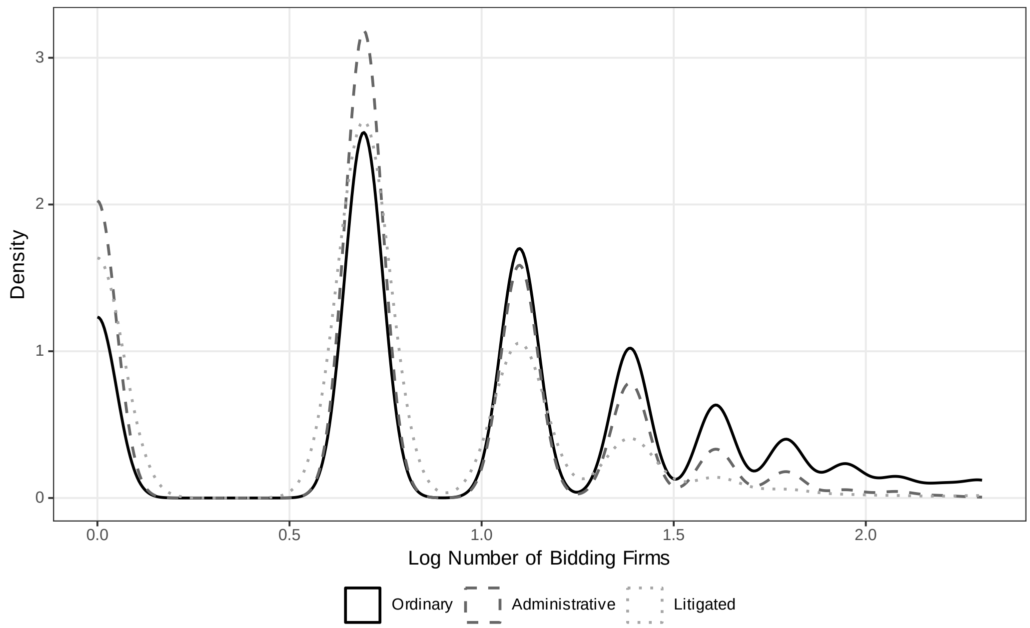


Figure A.6: Distribution of Log Number of Bidding Firms by Purchase Type

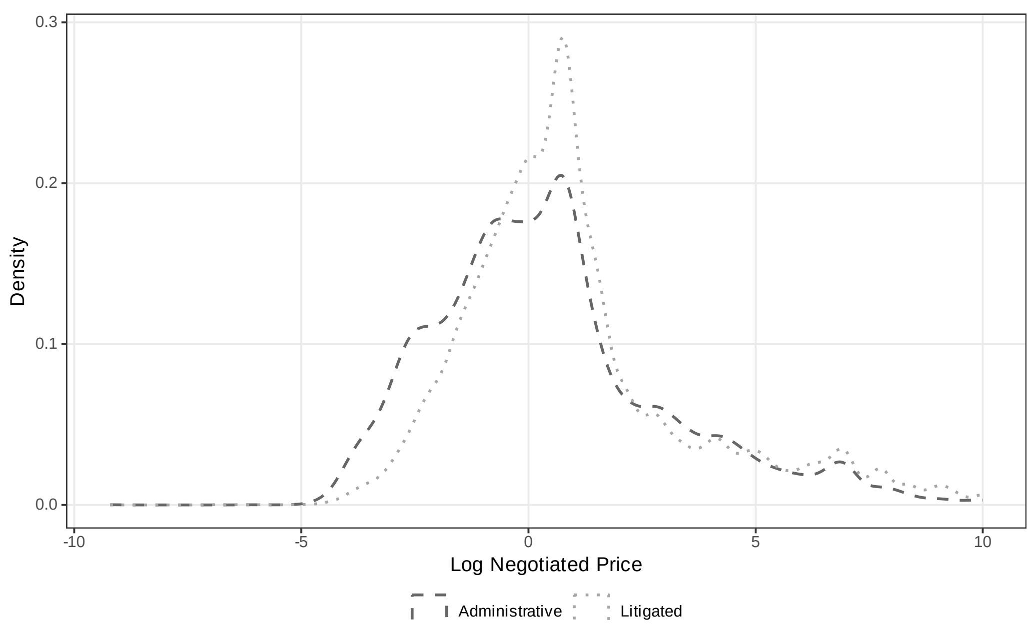


Figure A.7: Distribution of Log Negotiated Price: Administrative vs. Litigated (Urgent Only)

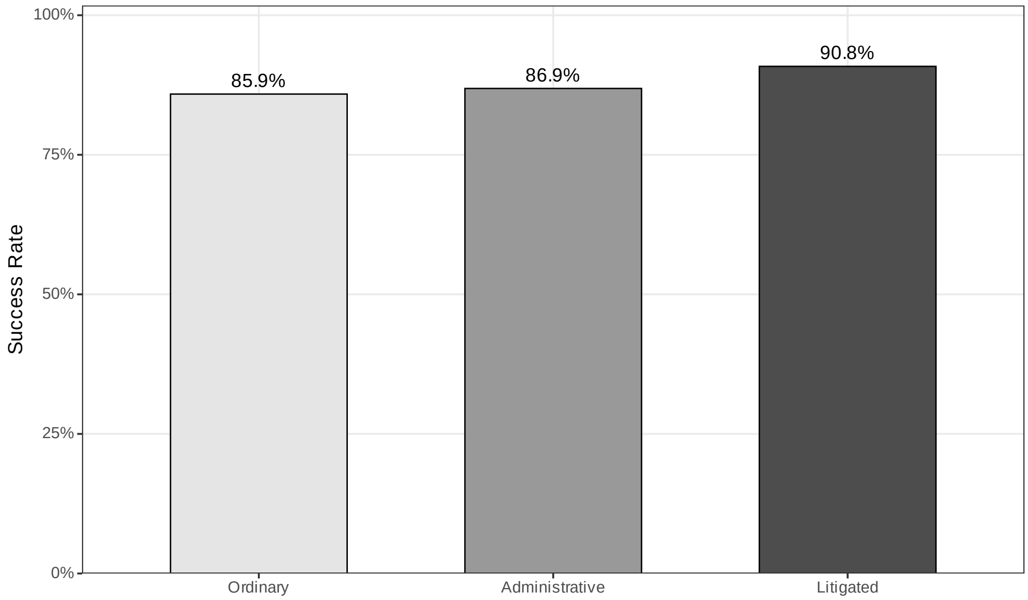


Figure A.8: Tender Success Rate by Purchase Type

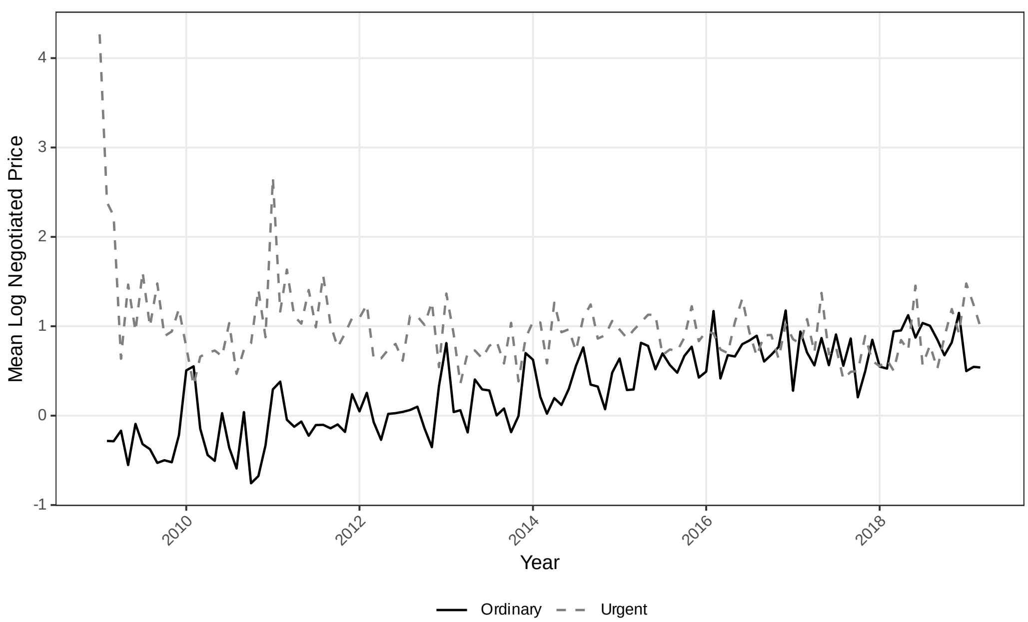


Figure A.9: Mean Log Negotiated Price Over Time: Ordinary vs. Urgent

A.4 Administrative Demand: Descriptive Evidence

This section presents geographic descriptive evidence on the distribution of administrative demand across municipalities in São Paulo, complementing the litigation maps in Section 2. Administrative purchases—urgent procurements initiated through internal government requests rather than court orders—share the same planning constraints as litigated purchases but carry no sanctions for procurement failure. The spatial patterns documented below confirm that administrative and litigated demand originate from similar geographic areas, supporting the institutional comparability that underlies the "under the gun" identification strategy.

Figure A.10 maps administrative purchases per 1,000 inhabitants across municipalities. The geographic distribution closely mirrors the litigation pattern shown in Figure A.1, with higher per capita rates in municipalities that host PBUs and in the interior of the state. This spatial overlap is consistent with both channels responding to the same underlying health needs.

Figure A.11 shows the location of PBUs that process administrative purchases, with marker size proportional to the volume of transactions. The distribution mirrors the PBU map for litigated purchases (Figure A.2), confirming that the same institutional infrastructure handles both procurement channels.

Figure A.12 provides a visual comparison of administrative and litigated purchases, highlighting that the procurement process is identical across both channels—the only distinguishing feature is the threat of judicial sanctions

Figure A.10: Administrative Purchases per 1,000 Inhabitants Across Municipalities in São Paulo (2009–2019)

Source: Authors' elaboration based on BEC procurement data and IBGE population estimates.

in litigated cases. This institutional parallel is the foundation for the "under the gun" identification.

Figure A.13 maps the total number of administrative purchases by municipality. As with litigation, administrative demand is concentrated in a small number of urban centers, particularly the São Paulo metropolitan area and regional hubs in the state's interior.

Figure A.14 shows the share of administrative purchases as a proportion of total purchases (administrative plus litigated) by municipality. The substantial variation across municipalities suggests that the relative importance of administrative versus litigated demand reflects local institutional factors—

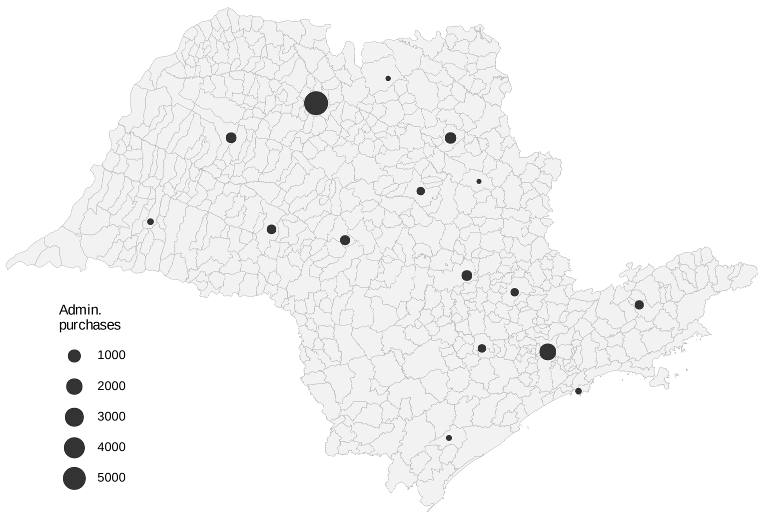


Figure A.11: PBUs Handling Administrative Purchases (Size $\propto$ Volume)
Source: Authors' elaboration based on SES/SP administrative records.

such as the presence of administrative request mechanisms and the propensity of local courts to grant injunctions—rather than a uniform statewide pattern.

	Administrative	Litigated
Origin	SES/SP request	Court order
Source of funds	Diverted budget	Diverted budget
Quantity	Small	Small
Delivery time	Short	Short
Threat of punishment	None	Potential punishment

Figure A.12: Comparison of Administrative and Litigated Purchases
Source: Authors' elaboration.

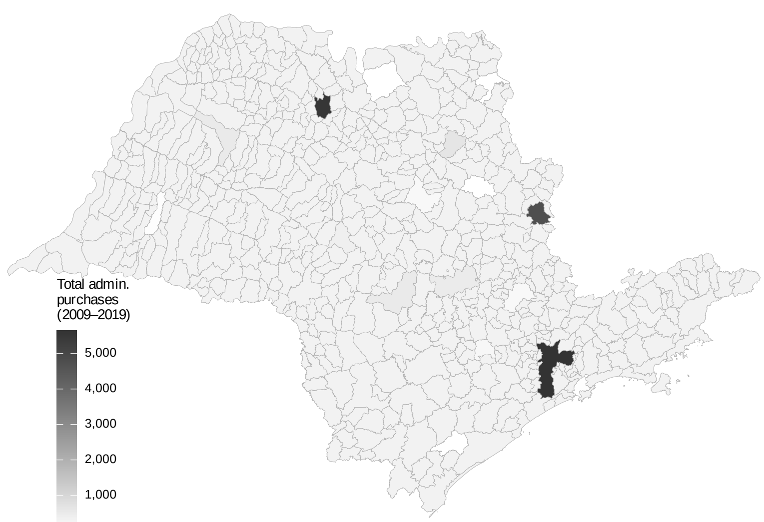


Figure A.13: Total Administrative Purchases by Municipality in São Paulo (2009–2019)
Source: Authors' elaboration based on BEC procurement data.

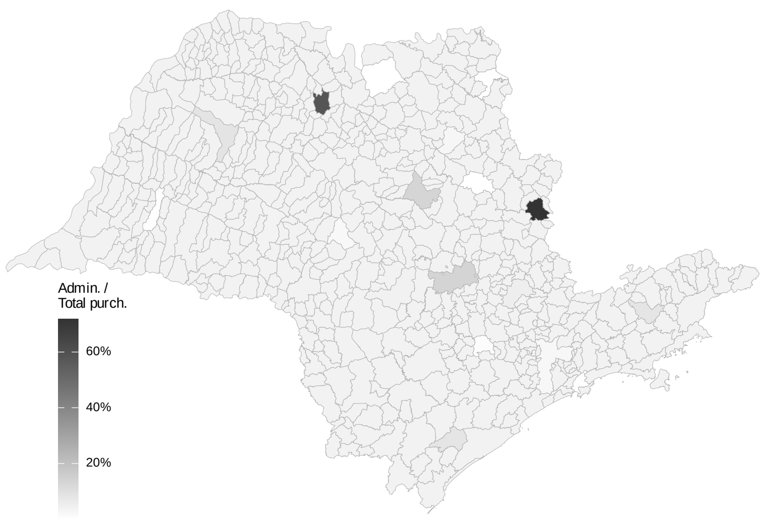
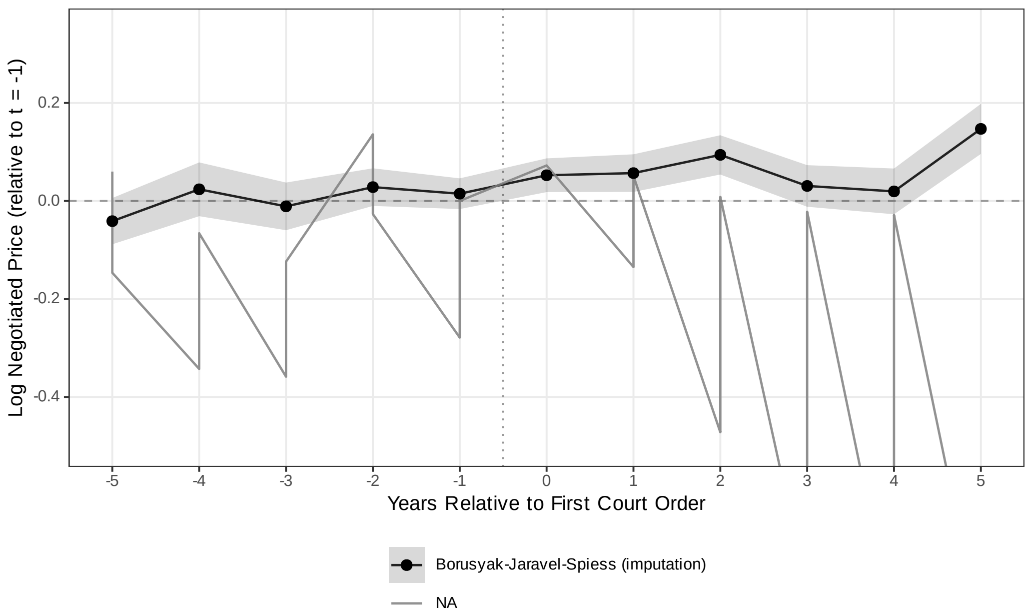


Figure A.14: Administrative Purchases as Share of Total Urgent Purchases by Municipality
Source: Authors' elaboration based on BEC procurement data.

A.5 Event Study: First Court Order (Honest DiD)

We re-estimate the dynamic response of log negotiated price (and log quantity) to the first court order for each item using two contemporary estimators: the imputation estimator of Borusyak et al. (2024) (BJS) and the ATT(g,t) estimator of Callaway and Sant'Anna (2021) (CS). The panel is item-year, winner-only; treatment is the calendar year of the first litigated purchase for each item; the never-litigated set is the CS control group. BJS pre-period coefficients are economically small (< 0.041 in absolute value); post-period coefficients rise from $+0.052$ at $t = 0$ to $+0.147$ at $t = +5$. Within the same item, order quantity drops sharply at the year of the first court order (-27.8% at $t = 0$) and recovers slowly (-0.3% at $t = +5$), tracing the C1 fragmentation channel in real time.

BJS is our preferred honest specification because it fits the counterfactual on pre-treatment observations of eventually-treated items, avoiding the selection problem of the CS never-treated control. The cross-sectional FE estimate in the main text is the primary object; the BJS event-study is supportive dynamic evidence.



referred (shaded 95% CI). CS with never-treated control is noisier because never-litigated items are a systematically different comparison group.

Figure A.15: Log negotiated price around first court order: honest-DiD estimators. *Notes:* BJS shown with shaded 95% CI; CS with never-treated control; TWFE as a pre-reform benchmark. CS is noisier because never-litigated items are systematically different (routine, well-stocked).

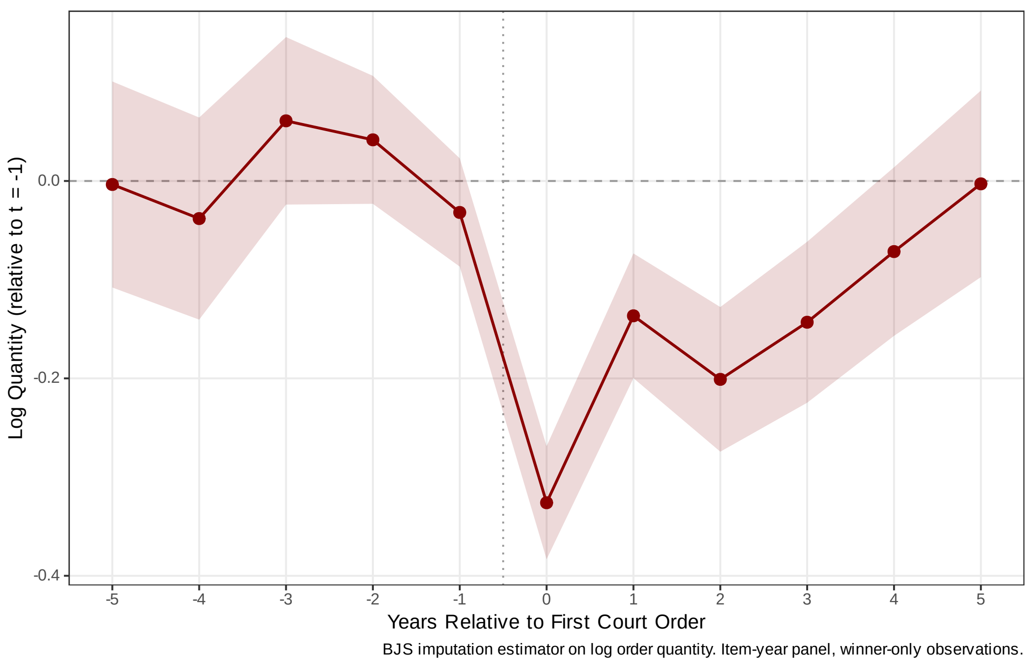


Figure A.16: Log order quantity around first court order: BJS imputation.

A.6 Heterogeneity

We examine heterogeneity in the urgency premium along four dimensions. Tables A.14–A.17 report results from split-sample regressions and interaction models using our preferred specification (item + year + PBU FE).

Table A.14: Heterogeneity: SUS Component

	Split Sample		Interaction
	Specialized	Basic SUS	Full Sample
	(1)	(2)	(3)
Urgent Purchase	0.005	0.030*	-0.073
	(0.046)	(0.016)	(0.097)
Urgent × Basic SUS			0.159
			(0.115)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	45,048	151,835	196,883
Within R^2	0.000	0.000	0.001

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** p<0.01, ** p<0.05, * p<0.1.

SUS component (Table A.14).. The urgency premium on negotiated prices is concentrated among items in the basic SUS component (common medications): the coefficient is 0.030 ($p < 0.10$) for basic items and an insignificant 0.005 for specialized items, suggesting that procurement disruption is more consequential for items where established supply chains normally deliver better prices.

Time period (Table A.15).. The urgency premium is larger in the later period (2014–2019) than in the earlier period (2009–2013), consistent with the

Table A.15: Heterogeneity: Time Period

	Split Sample		Interaction
	Early	Late (2014+)	Full Sample
	(1)	(2)	(3)
Urgent Purchase	0.074***	0.045**	0.259***
	(0.025)	(0.018)	(0.035)
Urgent × Late Period			-0.339***
			(0.043)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	82,658	113,936	196,883
Within R^2	0.000	0.000	0.005

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** p<0.01, ** p<0.05, * p<0.1.

growing volume and complexity of health litigation over time.

Market competition (Table A.16).. The urgency premium is three times larger in competitive markets (0.143 vs. 0.046), with a significant interaction (0.226, $p < 0.01$). In concentrated markets, fewer bidders leave less room for urgency to erode bargaining power; in competitive markets, the loss of planning time has the largest marginal impact.

PBU size (Table A.17).. The urgency premium is slightly larger for purchases made by smaller PBUs (below-median transaction volume), consistent with smaller units having less experience managing urgent procurement.

Table A.16: Heterogeneity: Market Competition

	Split Sample		Interaction
	Low Comp.	High Comp.	Full Sample
	(1)	(2)	(3)
Urgent Purchase	0.046***	0.143***	0.026
	(0.016)	(0.034)	(0.018)
Urgent × High Competition			0.226***
			(0.044)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	125,950	61,313	187,264
Within R^2	0.000	0.001	0.001

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** p<0.01, ** p<0.05, * p<0.1.

Table A.17: Heterogeneity: PBU Size

	Split Sample		Interaction
	Small PBU	Large PBU	Full Sample
	(1)	(2)	(3)
Urgent Purchase	0.023	0.062***	-0.000
	(0.026)	(0.018)	(0.049)
Urgent × Large PBU			0.070
			(0.054)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	31,902	164,511	196,883
Within R^2	0.000	0.000	0.000

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** p<0.01, ** p<0.05, * p<0.1.

A.7 Regex Classifier Validation

The purchase-type indicator is constructed by a regular-expression algorithm applied to public tender notices, which by Brazilian procurement law must reference any underlying court order. To document classifier accuracy, we draw a stratified random sample of 500 tender-notice subjects (approximately 167 per predicted class) and hand-label each as ordinary, administrative, or litigated. Hand-labeling of the stratified sample is in progress; F1 diagnostics will be reported in the replication archive accompanying this paper. The regex's literal-reference design implies misclassification, if any, is near-symmetric across the three classes: the algorithm relies on textual markers that are present or absent independently of the price outcomes we estimate, so misclassification attenuates rather than biases our estimates.[7]

A.8 Selection Bounds, Reconciliation, and Diagnostic Robustness

This section collects the selection-bound, reconciliation, and diagnostic robustness exhibits that support the main-text claims in Sections 4 and 5.

Manski-Lee bounds on the UTG.. Table A.18 reports the bounded UTG coefficient under monotone selection. Trimming admin observations within item×year×PBU strata where the admin sample exceeds the litigated count yields a bounded gap in $[15.9\%, 21.1\%]$, against a naive coefficient of 29.5%.

[7] The stratified sample is reproduced from `output/validation/validation_sample.csv`, generated by `analysis/33_regex_validation_sample.R`; F1 statistics are emitted by `34_regex_validation_f1.R` when the labeled CSV is finalized.

Table A.18: Manski-Lee bounds on the under-the-gun coefficient.

	log negotiated price		implied
Specification	Coef. on Admin	SE	Lit-vs-Admin (%)
Naive UTG (item + year + PBU FE)	-0.259	0.092	29.530
Lee lower bound (admin top-trim)	-0.192	0.095	15.924
Lee upper bound (admin bottom-trim)	-0.148	0.097	21.123

Trimming applied within item*times*year*times*PBU strata where the admin sample exceeds the li

Average trimming intensity is 26.9% of admin observations across 12,038 of 44,654 strata.

Heckman selection model (non-identified).. Table A.19 reports a parametric Heckman correction with the same first-stage probit (pre-period log reference price, mean quantity, mean number of bidders, and order count as the exclusion restriction; SUS-formulary indicator). The probit confirms negative loadings on pre-period reference price and SUS status. The inverse Mills ratio is collinear with the item, year, and PBU fixed effects in the second stage, so the resulting Admin coefficient (0.817, SE 4.278) is uninformative. The Lee bound is the primary selection-corrected object.

Both-types-cell representativeness.. Table A.20 compares the 5,581 item×year-month cells that contain both administrative and litigated urgent purchases (the within-cell variation that identifies the tightest specification) with the 11,972 singleton cells. Both-types cells systematically have lower mean log reference and negotiated prices, larger quantities, slightly fewer bidders, and a higher SUS-formulary share. Bounded UTG estimates therefore speak to

Table A.19: Heckman selection-corrected UTG estimate.

Term	Estimate
First stage: pre-period log ref price	-0.051
First stage: pre-period mean qty	0.000
First stage: pre-period mean firms	0.065
First stage: pre-period n orders (excl)	0.000
First stage: SUS formulary	-0.265
Second stage: Admin coef	0.817
Second stage: inverse Mills ratio	-0.653
Second stage: implied lit-over-admin (%)	-55.841
Heckit ML: ρ	1.000

First stage: probit of admin on pre-period item characteristics. pre_n_orders is excluded from t

the more liquid, formulary-anchored items in the urgent panel; we read this as a generalizability boundary, not as a threat to within-cell identification.

Wild cluster bootstrap.. Table A.21 reports Rademacher wild cluster bootstrap inference on the UTG coefficient (999 replicates, PBU clustering, restricted-null residuals).[8] The preferred-FE Admin coefficient is significant under both asymptotic and bootstrap inference ($p_{\text{boot}} = 0.0080$, 95% CI in log-points $[-0.469, -0.039]$); the item×year-month FE specification is borderline ($p_{\text{boot}} = 0.0390$), as expected given the thin within-cell sample.

Rambachan-Roth honest sensitivity on BJS event-study.. Figure A.17 reports the Borusyak-Jaravel-Spiess imputation event-study coefficients with a Rambachan-Roth honest sensitivity overlay. Pre-period maximum absolute

[8] Manual Rademacher implementation; `fwildclusterboot` unavailable for R 4.5. See `analysis/44_wild_bootstrap.R`.

Table A.20: Representativeness of both-types cells: cells with both administrative and litigated urgent purchases (5,581 cells, 31.8%) vs. singleton cells (11,972).

Variable	Both-types cells	Singleton cells
Log reference price	1.233	1.761
Log negotiated price	0.703	1.326
Log quantity	6.560	6.324
N. bidding firms	2.181	2.404
SUS formulary share	0.867	0.725
Reverse-auction (pregao) share	0.955	0.952

Means computed at the observation level within each cell type. p-values from two-sample t-tests.

Table A.21: Wild cluster bootstrap inference on the UTG coefficient.

Specification	$\hat{\beta}$	Asy. SE	t	p_{boot}	95% CI (log-points)
Preferred (item + year + PBU)	−0.259	0.092	−2.80	0.008	[−0.469, −0.039]
Tightest (item×year-month + PBU)	−0.283	0.116	−2.44	0.039	[−0.549, −0.001]

Notes: Rademacher wild cluster bootstrap on the Admin coefficient with $B = 999$ replicates and restricted-null residuals; clustering at the PBU level. Negative coefficients mean litigated purchases are more expensive than administrative ones, in log-points. Source: `analysis/44_wild_bootstrap.R`.

coefficient is 0.041; the $t = 0$ effect is 0.052 (SE 0.018); the breakdown linear-extrapolation parameter at which the $t = 0$ confidence interval just contains zero is $M = 0.018$. The $t = 0$ effect is significant under parallel trends but does not survive the linear extrapolation of the observed pre-period maximum—we honor this in the main text by treating the cross-sectional fixed-effects estimate as primary and the BJS event study as supportive, not as a stand-alone identification claim.

Reconciliation table.. Table A.22 decomposes the observed log-price gap (admin minus litigated) into the mechanical bulk-discount channel (−32.8%, IC

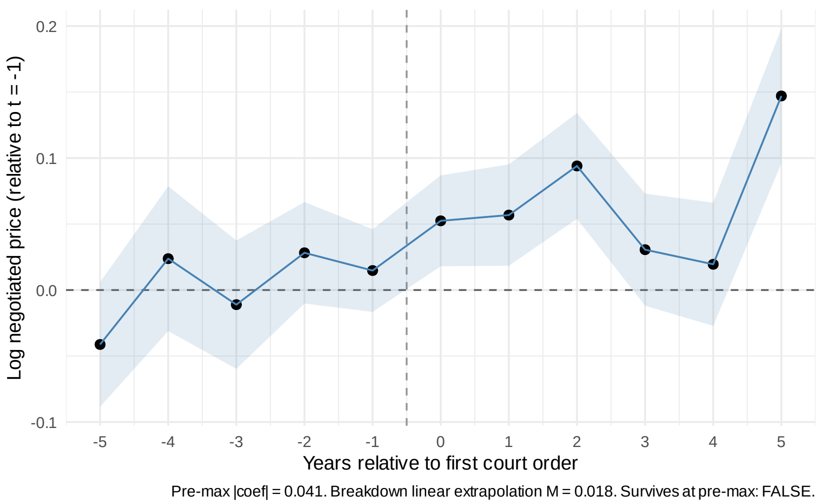


Figure A.17: BJS event study around first court order with Rambachan-Roth sensitivity diagnostic.

[-51.2%, -5.2%]), the within firm-buyer-item offset (3.5%, IC [-6.6%, 11.0%]), and the supplier composition residual (10.9%, IC [-16.0%, 39.6%]). Within-firm pricing is identified and zero; bulk-discount is identified and large; the composition residual is imprecisely identified by elimination.

Table A.22: UTG reconciliation: observed log-price gap decomposed into mechanical bulk-discount prediction, within firm-buyer-item offset, and supplier composition residual. 1,206 triples observed in both regimes.

Component	Log-points (admin minus lit)
Observed UTG (admin minus lit, log price)	-0.259
Mechanical C1: bulk-discount x qty gap	-0.397
Within firm-buyer-item offset	0.035
Supplier composition residual	0.104

Identity: observed UTG = mechanical C1 + within-firm offset + composition residual. Within-fi

Table A.23: Selection-bias-corrected welfare bound on the procurement cost of judicial enforcement.

Component	Value
Bounded UTG (Lee midpoint)	18.5%
Lee lower bound	15.9%
Lee upper bound	21.1%
Admissibility share (calibration)	50%
Annual litigated spend, SP	\$300 M / yr
Welfare bound (Lee midpoint x admissibility x spend)	\$27.8 M / yr
Welfare bound (Lee lower x admissibility x spend)	\$23.9 M / yr
Welfare bound (Lee upper x admissibility x spend)	\$31.7 M / yr

Bounded UTG: Manski-Lee bounds from Table A.18. Admissibility share: calibration anchor for

Welfare bound.. Table A.23 reports the selection-bias-corrected welfare bound. Applying the Lee-midpoint UTG of 18.5% to \$300 M of annual litigated spending in São Paulo, with an admissibility-share calibration of 50.0%, yields \$27.8 M per year on the per-unit-price margin. The Lee-endpoint range is $[\$23.9\ M, \$31.7\ M]$. This is a per-unit-price bound and excludes welfare losses from constrained order sizes, reduced bidder participation, and officials' search time diverted from ordinary procurement; the headline number understates the total cost.